

Ultralow-Tensile Strain Enables Exciton Funneling and Energy Transfer to Boost MoSe₂ Photoluminescence Quantum Yield

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Strain engineering is a powerful route for controlling the exciton dynamics in van der Waals (vdW) heterostructures (HSs). The interlayer energy transfer (ET) process is another key factor in controlling the photocarrier relaxation pathways in vdW HSs. In this work, we combine these two processes to achieve an ~ 8 -fold enhancement to the relative photoluminescence (PL) quantum yield (QY) in a HS formed from monolayers of ReS₂ and MoSe₂, separated by a thin hBN interlayer, placed onto an hBN bubble. We achieve this enhancement by applying only $\sim 0.1\%$ biaxial tensile strain, which results in efficient exciton funneling and an increased transition dipole moment. Our experimental data are supported by first-principles density-functional theory and coherent transfer-matrix method calculations, ruling out optical interference as the dominant origin of the enhancement. This work provides an innovative route for enhancing the PL QY of vdW materials via interplay between the tensile strain and the ET process.

I. INTRODUCTION

Mechanical strain has emerged as an effective tool for tailoring the electronic, optical, and excitonic properties of two-dimensional (2D) van der Waals (vdW) semiconductors [1–3]. In monolayer (ML) transition metal dichalcogenides (TMDCs), local strain is commonly introduced using bubbles, wrinkles, nanopillars, or patterned substrates [4–8]. In particular, strain generated by bubbles creates localized potential landscapes that promote exciton funneling and modify light emission — providing an effective route toward tunable nanoscale optoelectronic and quantum photonic devices [9–11]. When different TMDC MLs are vertically stacked to form heterostructures (HSs), strain also influences the interlayer coupling [12–14].

Interlayer energy transfer (ET) process, particularly non-radiative Förster resonance ET provides an efficient pathway to control the exciton dynamics in vdW HSs without relying on the interlayer charge transfer (CT) process [15–19]. In type-II TMDC HSs, ET competes with CT and strongly depends on the donor-acceptor spectral overlap, and interlayer separation [15, 19–22]. A recent experimental study has shown the effect of compressive strain on the ET process by placing a TMDC HS inside a diamond anvil cell [23]. However, the influence of the small tensile strain that inevitably arises from

bubble formation during the HS fabrication remains experimentally unexplored. Thus, understanding how weak local tensile strain modifies the interlayer exciton dynamics is timely and practically relevant. In this work, we demonstrate that only $\sim 0.1\%$ biaxial tensile strain produces an ~ 8 -fold enhancement of the relative MoSe₂ photoluminescence (PL) quantum yield (QY) in an all-ML ReS₂/hBN/MoSe₂ HS, by placing the entire stack onto a hBN bubble/sapphire substrate. The thin (~ 12 nm) hBN spacer suppresses the CT process, while enabling the long-range dipole–dipole ET [21]. The strain simultaneously drives the exciton funneling and increases the transition dipole moments (TDMs) of both MLs — boosting the ET efficiency. First-principles calculations and transfer-matrix modeling (TMM) support the experimental findings and rule out the possibility of optical interference as the major contributor of the observed enhancement. Importantly, the present results are obtained in a purely ML HS; increasing the optical absorption of the donor layer is therefore expected to yield even larger gains in MoSe₂ emission. Our findings also promise to further increase the PL quantum yield (QY) of any TMDC HS by applying a higher strain in a controllable way — ideal for real vdW optoelectronic device applications.

II. RESULTS AND DISCUSSION

The investigated HS is assembled layer-by-layer on a specially prepared hBN bubble/sapphire substrate,

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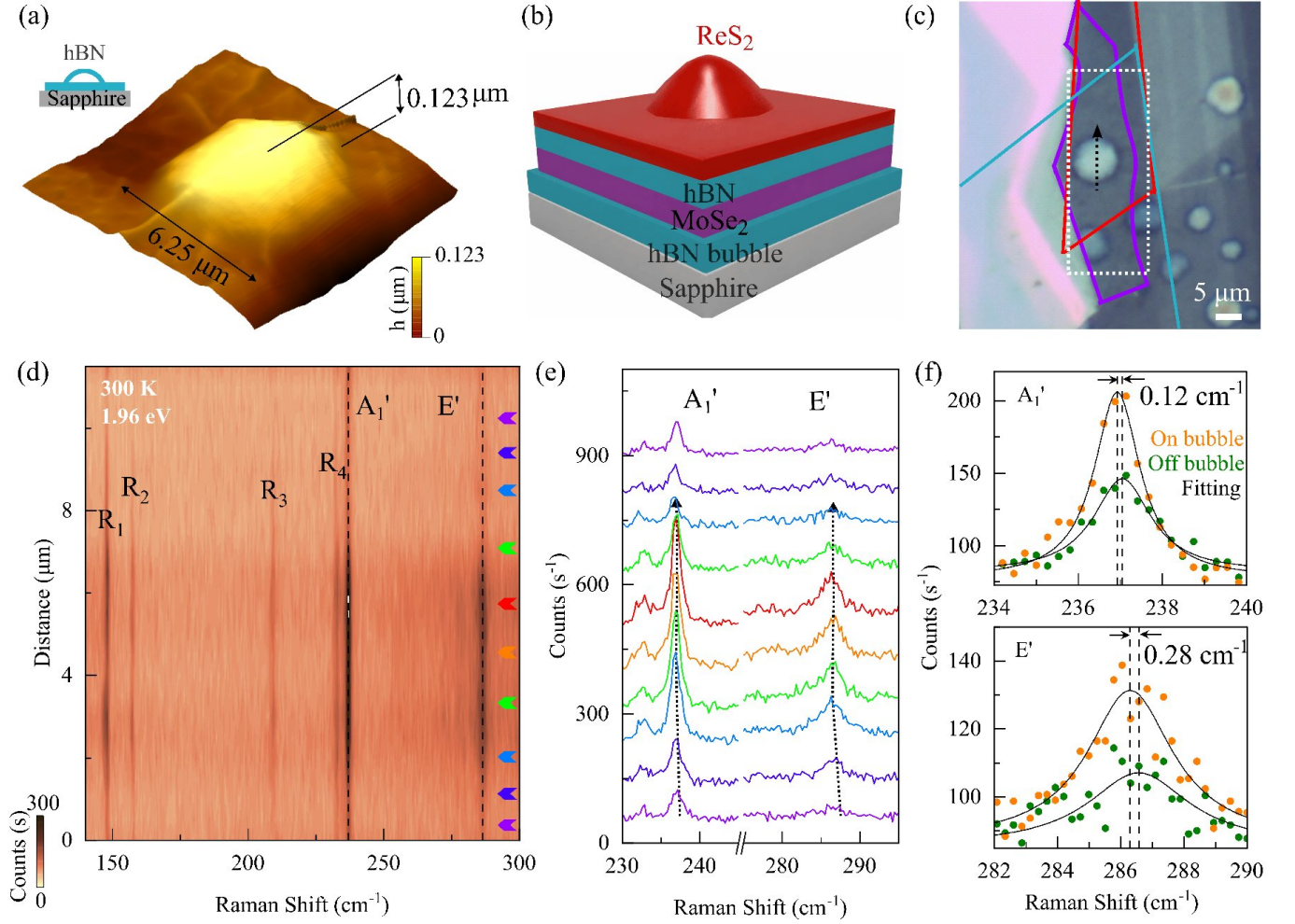


FIG. 1. (a) Three-dimensional AFM image of the hBN bubble on the sapphire substrate, showing a height of $\sim 0.12 \mu\text{m}$ and base diameter of $\sim 6.25 \mu\text{m}$. (b) Schematic illustration of the HS cross-section (from top): ML ReS_2 , interlayer hBN, ML MoSe_2 and the bottom hBN bubble/sapphire substrate. (c) Optical image of the sample. Colored outlines correspond to the layers shown in (b). (d) RT Raman intensity map along the black dotted arrow shown in (c) measured using 1.96 eV excitation. Overall, Raman intensity on the bubble is significantly higher as compared to the outside of the bubble. (e) Selected Raman spectra from different positions along the line scan shown in (d) — showing small shifts in MoSe_2 A_1' and E' modes at the bubble apex. Vertical offsets are added to the spectra for a better visual representation. (f) Top and bottom panels show zoomed in MoSe_2 Raman peaks — then Lorentzian fits are applied to determine the peak shifts from the on-bubble (strained) and off-bubble (unstrained) positions. The extracted redshifts from the unstrained to strained region are $|\Delta\omega_{A_1'}| = 0.12 \text{ cm}^{-1}$ and $|\Delta\omega_{E'}| = 0.28 \text{ cm}^{-1}$.

which provides a well-defined platform for strain engineering [24]. A detailed discussion of the hBN growth, bubble formation and HS fabrication is provided in the methods section. Figure 1(a) shows a three-dimensional atomic force microscopy (AFM) image of the hBN bubble on the sapphire substrate after transferring ML MoSe_2 . The bubble exhibits a height of $0.12 \mu\text{m}$ and a base diameter of $6.25 \mu\text{m}$. Following the bubble profile reported by Khestanova *et al.* [25] and the geometrical approach for estimating strain from the bubble morphology [26, 27], the geometrical strain is estimated to be $\sim 0.3\%$ [calculations shown in Section S1 of the Supplementary Information (SI)]. This estimated value represents an upper bound, since elastic relaxation during bubble formation

lowers the actual strain transferred to the MLs. The actual strain present in the transferred HS is therefore further verified independently by Raman and PL measurements as discussed in the following section. Figure 1(b) presents a schematic cross-sectional view of the sample and Fig. 1(c) shows the optical image of the sample, in which the bubbles appear as bright spots and the layer boundaries are color-coded consistently with the schematic. The thickness of the interlayer hBN of $\sim 12 \text{ nm}$ (AFM height profile is shown in SI Section S2) is chosen to completely suppress the CT process [21].

To characterize the strain distribution across the bubble, room temperature (RT) Raman line scans under 1.96 eV excitation energy are performed [Fig. 1(d)] from

the flat HS region towards the apex of the bubble along the direction indicated by the black dotted arrow in Fig. 1(c). Four ReS_2 Raman-active modes (R_1 – R_4) are identified at 148.03, 157.39, 209.18, and 232.85 cm^{-1} . The MoSe_2 A'_1 and E' modes are observed at 236.92 and 286.29 cm^{-1} , respectively. Individual Raman spectra are shown in SI Section S3. The map reveals a clear enhancement of Raman intensity in the strained region for all modes, indicating an efficient ET process [28]. For a quantitative analysis of the strain magnitude and character, we focus on the MoSe_2 A'_1 and E' modes, due to their higher symmetry compared with the low-symmetry ReS_2 modes. Representative MoSe_2 Raman modes at several positions along the Raman map are presented in Fig. 1(e). Both modes exhibit a systematic increase in intensity and a small but measurable redshift when moving from the flat region to the bubble apex. The E' mode shows a larger shift than the A'_1 mode, consistent with its greater sensitivity to strain [29, 30]. The nature of the strain is first assessed *via* the intensity ratio $I_{E'}/I_{A'_1}$. In the unstrained region, this ratio is 0.49 and increases to 0.57 over the bubble — demonstrating the presence of biaxial tensile strain [29].

For a numerical determination of the MoSe_2 Raman shifts, Lorentzian fits are applied to spectra extracted from the on-bubble (strained) and off-bubble (unstrained) positions, as shown in Fig. 1(f). The extracted redshifts from the unstrained to strained region are $|\Delta\omega_{A'_1}| = 0.12 \text{ cm}^{-1}$ and $|\Delta\omega_{E'}| = 0.28 \text{ cm}^{-1}$. These redshifts are consistent with the phonon softening due to the elongation of the Mo–Se bonds under a tensile strain, which reduces the interatomic force constants [31]. The strain magnitude is estimated using the biaxial Raman shift coefficients reported for MoSe_2 in Ref. [29] (details are in SI Section S1). The calculated data shows a consistent picture of a weak local biaxial tensile strain between ~ 0.09 – 0.12% , matching well with earlier reports [32]. An independent estimation derived from the PL peak redshift taken along the arrow in Fig. 1(c) (see SI Section S4) indicates a tensile strain $\sim 0.06\%$. This slight difference in the strain value is expected due to the high-sensitivity of the Raman modes for lattice distortions [29]. Due to the limited spatial resolution of the optical setup, the discussion is restricted to the flat (off-bubble) region and the bubble apex; intermediate sidewall regions are not analyzed.

To probe the effect of this localized strain on the optical response, RT PL mapping is performed under 2.40 eV excitation. In order to decouple any optical interference effects arising from the hBN bubble/sapphire cavity, PL measurements are first carried out on the partial stack (hBN/ MoSe_2 /hBN bubble/sapphire) before transferring the top ReS_2 layer [Fig. 2(a)]. The map reveals a localized enhancement of the MoSe_2 emission at the bubble apex. This enhancement is attributed to the strain-induced funneling effect [6, 7]. The tensile strain generates a spatial variation in the MoSe_2 optical-transition energy, as indicated by the PL energy redshift (SI Sec-

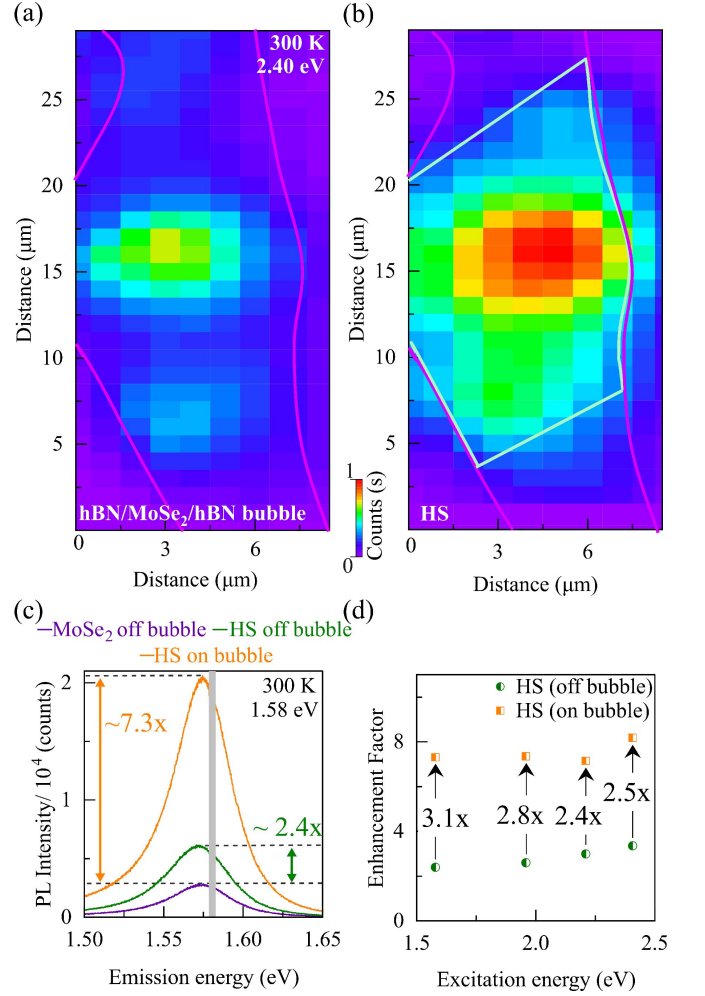


FIG. 2. (a) Spatially resolved RT PL intensity map under 2.40 eV excitation from the white dotted area marked in Fig. 1(c) prior to the transfer of top ReS_2 layer. The marked area indicates the ML MoSe_2 region. MoSe_2 emission is stronger in the bubble region than in the surrounding flat region. (b) PL intensity map of the same region under the same experimental conditions, after placing the top ReS_2 layer. HS area at the bubble apex shows the maximum PL intensity. (c) RT PL spectra measured under 1.58 eV excitation, from ML MoSe_2 , on bubble and off bubble HS areas show $\sim 7.3\times$ and $\sim 2.4\times$ enhancement, respectively. The gray shaded line indicates the spectral range blocked by the Bragg filter to suppress the leaking from the excitation laser. (d) PL enhancement factor as a function of the excitation energy. The maximum PL enhancement factor between the HS regions of $\sim 3\times$ was observed under 1.58 eV excitation.

tion S4). As a result, photoexcited excitons drift toward the region of maximum tensile strain, increasing the local exciton population and consequently enhancing PL emission from MoSe_2 [6, 7]. After the transfer of the top ML ReS_2 to complete the HS, the PL map of the same region is re-acquired [Fig. 2(b)]. The PL intensity map clearly shows that the HS emission in the strained region is significantly enhanced. Quantitative PL spectra

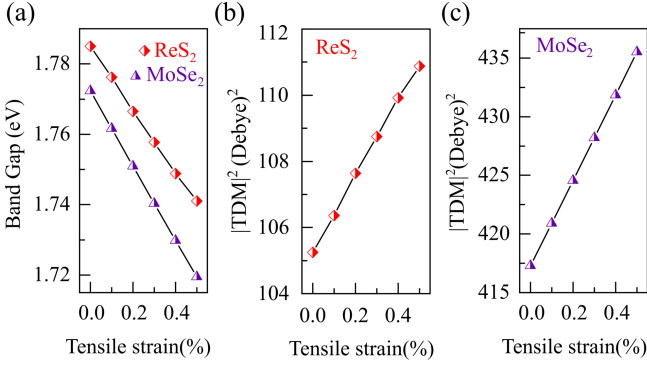


FIG. 3. (a) First-principles DFT calculations of the electronic band structures of both TMDCs showing a continuous reduction of the bandgaps with an increasing biaxial tensile strain. The solid lines are guides to the eye. ReS₂ always maintains a higher bandgap than MoSe₂ throughout this range. (b–c) Calculated $|TDM|^2$ for MLs of ReS₂ and MoSe₂, respectively, as a function of biaxial tensile strain. $|TDM|^2$ shows a steady increase with the tensile strain.

under 1.58 eV excitation comparing the three regions are shown in Fig. 2(c). This excitation energy is chosen to selectively populate the A exciton of MoSe₂, reducing the inter-/intravalley carrier scattering to increase the ET efficiency. The MoSe₂ emission in the unstrained HS is enhanced by a factor of $\sim 2.4\times$ relative to the ML MoSe₂ layer [Fig. 2(c)] — demonstrating an ET process from the ReS₂ to the MoSe₂ layer [21]. In the strained HS, this enhancement factor increases to $\sim 7.3\times$ [Fig. 2(c)], demonstrating that localized tensile strain enhances the ET efficiency. The PL enhancement factor here is defined as the integrated MoSe₂ PL intensity in the HS normalized to the ML MoSe₂ layer. The reported enhancement factors are corrected for the attenuation of the excitation beam caused by the absorption of the top ML ReS₂ in HS (SI Section S5). The possible contributions of exciton funneling, interlayer ET and optical-interference effects are examined in the later sections. We would like to point out that under the present experimental conditions, no distinct ReS₂ PL emission is resolved, and the detected HS emission is therefore dominated by the MoSe₂ spectral contribution [21].

Additional PL measurements are performed using excitation energies of 1.96, 2.21 and 2.40 eV (SI Section S5 and Fig. 2(d)). The maximum HS PL enhancement factor of $\sim 3\times$ under 1.58 eV excitation in Fig. 2(d) is attributed to the combination of exciton funneling, reduced inter-/intravalley carrier scattering, and suppression of non-radiative losses — resulting in maximizing the ET efficiency. A similar enhancement of the HS emission is consistently observed in two supporting samples, confirming the reproducibility of the observed effect (SI Section S6). After correcting for the optical absorption at each excitation energy, the relative PL QY of the strained HS is enhanced by $\sim 8\times$, $7\times$, $7\times$, and $8\times$ (SI Section S7) compared to the ML MoSe₂ at excitation energies of 1.58,

1.96, 2.21, and 2.40 eV, respectively.

To gain theoretical insight into the enhanced MoSe₂ emission, first-principles electronic-structure calculations are performed for biaxially strained ML ReS₂ and ML MoSe₂ (details in SI Section S8). The evolution of the electronic direct gaps with biaxial tensile strain is summarized in Fig. 3(a), calculated at the band minima *i.e.*, Γ valley for ReS₂ and the K valley for MoSe₂ [33, 34]. For the experimentally relevant strain range, both materials exhibit an approximately linear reduction in bandgap with increasing tensile strain. This calculated trend is qualitatively consistent with the small PL redshift observed in the bubble region (SI Section S4), indicating that the biaxial tensile strain reduces the bandgap [33, 34]. The optical transition probability is proportional to the intralayer $|TDM|^2$ [19]. The calculated $|TDM|^2$ for the ReS₂ and MoSe₂ as a function of biaxial tensile strain are shown in Figs. 3(b) and (c), respectively (SI Section S8). The linear increase in $|TDM|^2$ for both MLs with the tensile strain, may help in enhancing the dipole-dipole coupling [19]. Due to a limited computational capability, only single-particle band structure is considered and a discussion related to the change in the excitonic binding energy as a function of strain is omitted.

A schematic illustration of the bandgap modulation and exciton funneling along the bubble is presented in Fig. 4(a), where at the apex of the bubble the TMDC bandgap shows the maximum reduction due to the maximum biaxial tensile strain [6, 7]. Figure 4(b) shows the proposed ET mechanism: gray and colored bands represent the unstrained and strained band minima, respectively. At the bubble apex, excitons accumulate *via* funneling and an increased TDM helps in an effective ET coupling from the ReS₂ to MoSe₂ layer. The slightly higher ReS₂ excitonic energy compared to MoSe₂ [Fig. 3a] [21], further increases the probability of dipolar coupling outside of the light cones. A combination of all these factors resulted in the observed HS PL enhancement at the bubble apex.

To probe the influence of strain on the exciton dynamics, RT time-resolved PL (TRPL) measurements are performed on isolated ML MoSe₂, and on both the HS regions using an excitation energy of 1.61 eV [Fig. 4(c)]. The decay curves are fitted with a biexponential decay function, yielding a fast (τ_1) and a slow (τ_2) decay component. τ_1 is associated with the intrinsic PL decay, whereas τ_2 reflects the slower nonradiative recombination processes [35]. We will thus focus only on τ_1 in our discussion. For isolated ML MoSe₂, $\tau_1 = 13.8 \pm 0.3$ ps. In the strained and unstrained HS regions, the corresponding values are 19.0 ± 0.3 and 29.9 ± 0.8 ps, respectively. The slower τ_1 in the unstrained HS as compare to ML MoSe₂ is a typical signature of an ET process [18, 19]. In comparison, a shorter τ_1 observed for the strained HS (but comparable with that of ML MoSe₂); indicating a faster exciton recombination under near-resonant excitation. This behavior is attributed to the strain-induced bandgap narrowing, which increases the exciton

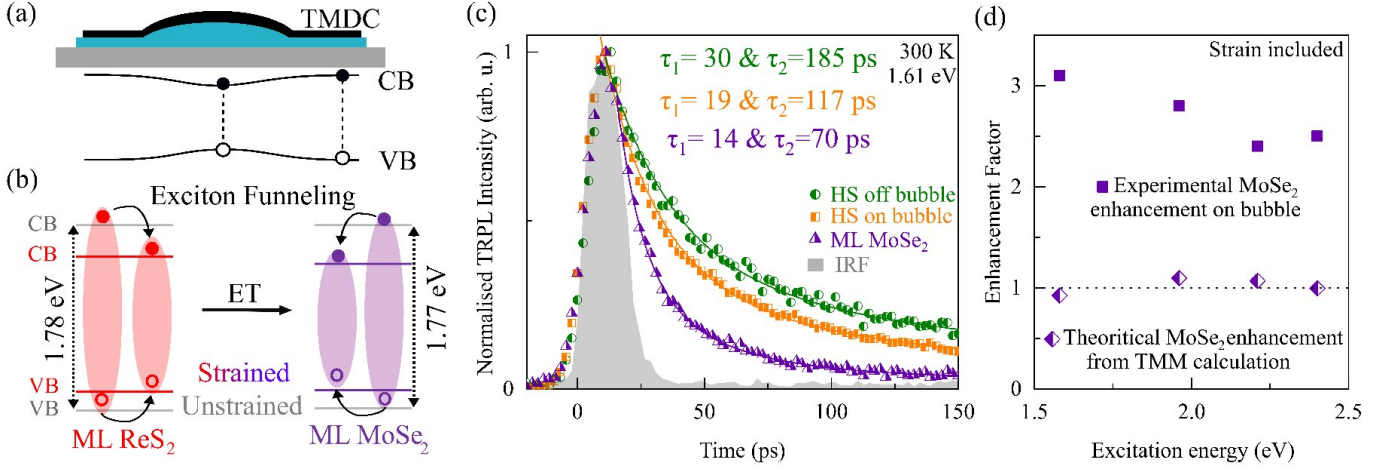


FIG. 4. (a,b) Schematic illustration of strain-induced bandgap modulation and exciton funneling and the proposed dipole-mediated ET pathway from ReS₂ to MoSe₂. (c) RT TRPL spectra acquired from the ML MoSe₂, and the strained and unstrained HS area under 1.61 eV excitation. The gray shaded region shows the instrument response function (IRF). The decay curves are fitted using a biexponential decay function to extract the fast (τ_1) and slow (τ_2) decay components. The strained HS exhibits a faster decay components than the unstrained HS. (d) Strain dependent TMM calculation of the MoSe₂ optical enhancement factor shows a near unity value for the entire excitation range. The adopted model shows that the optical interference effect from the bubble structure is insufficient to account for the measured PL enhancement.

funneling and enhances the radiative recombination *via* ET process. In contrast, under above-bandgap excitation at 2.27 eV (SI Section S9), the decay dynamics of the strained and unstrained HS become nearly identical. At these higher excitation energies, carrier relaxation through phonon-assisted inter-/intravalley scatterings dominate the exciton dynamics, largely masking the comparatively small influence of strain-induced exciton funneling. It is important to mention that, since the HS PL emission is dominated by the MoSe₂ signature, we assigned the HS TRPL spectra only for the MoSe₂ contribution. Resolving the ReS₂ emission from the TRPL data is beyond our instrumental resolution, a separate ultrafast study is required for that.

Finally, to evaluate whether the enhanced MoSe₂ emission could partly originate from the bubble-induced optical interference, the optical response of the HS is modeled using a coherent TMM method [36–38]. The model consists of an air/ML ReS₂/hBN/ML MoSe₂/hBN bubble/sapphire stack (details are provided in SI Sections S10-S12). Our model accounts for multiple reflections, optical interference effects, and absorption within the multilayer stack. The theoretically calculated MoSe₂ optical enhancement factor shown in Fig. 4(d) remains close to unity over the investigated excitation-energy range, indicating that the bubble geometry produces only a weak modulation of the optical absorption. A similar calculation for the ReS₂ layer, presented in Section S13 of the SI, leads to the same conclusion that optical interference is substantially smaller than the measured PL enhancement. Within the adopted model, the optical interference alone is therefore insufficient to account for the pronounced HS PL enhancement. The remaining enhancement is a result of an enhanced ET process due to

a combination of strain-modified exciton dynamics and exciton funneling. However, the individual contributions of these processes cannot be quantitatively separated by the present measurements.

III. CONCLUSIONS

In summary, we demonstrate that only $\sim 0.1\%$ biaxial tensile strain produces an ~ 8 -fold increase in the relative MoSe₂ PL QY in MLs ReS₂/hBN/MoSe₂ HS. The thin (~ 12 nm) hBN spacer suppresses the CT process. Raman and PL measurements consistently indicate weak tensile strain, while first-principles calculations reveal strain-induced bandgap narrowing and increased TDMs that strengthen dipole-dipole coupling. The strain gradient further promotes the exciton funneling toward the bubble apex. TMM calculation rules out optical interference as the dominant origin of the enhancement. Importantly, these results are obtained in a purely ML HS; increasing the optical absorption of the donor layer is therefore expected to enable even larger gains. Our findings establish small tensile strain as a practical and controllable route to engineer ET and PL QY in ultrathin vdW optoelectronic devices.

IV. METHODS

A. Sample Preparation

hBN Growth on Sapphire Substrate: The hBN layers were grown on 2-inch c-plane sapphire substrates with

a 0.3° offcut using metal-organic vapor phase epitaxy (MOVPE) in an Aixtron CCS 3×2 system equipped with an ARGUS thermal mapping system. Triethylboron and ammonia were used as boron and nitrogen precursors, respectively, while hydrogen served as the carrier gas. The growth was performed at 1400°C using a two-stage epitaxial growth process described in Ref. [39].

hBN Bubble Formation: Localized bubbles were generated by electron-beam exposure using an FEI Helios 600 Dual Beam system equipped with a Raith Elphy electron lithography setup. During the exposure, the electron beam current and acceleration voltage were maintained at 1 nA and 5 keV, respectively. The resulting bubbles are mechanically stable under ambient conditions and retain their geometry over extended periods. Previous study on analogous hBN bubbles reported tensile strains of approximately 0.1% [40], consistent with the strain determined in the present work from Raman and PL measurements.

HS Fabrication: hBN crystals were obtained from the National Institute for Materials Science (NIMS), Japan, while bulk MoSe_2 and ReS_2 crystals were purchased from 2D Semiconductors, USA. ML TMDCs and thin hBN flakes were mechanically exfoliated onto polydimethylsiloxane (PDMS) gel films. The layer thicknesses were confirmed by the optical microscope, PL and Raman spectroscopy measurements. The vertical stacking of the layers was carried out using a deterministic dry-transfer technique. After each layer is transferred, the samples were annealed inside an e-beam evaporator chamber under high vacuum ($\sim 7 \times 10^{-4}$ Torr) at 180°C for 3 h under argon flow of 50 sccm to improve the interlayer adhesion and interface quality.

B. Characterization

Optical Characterization: PL measurements were performed using Cobolt diode lasers with excitation energies of 1.58, 1.96, 2.21, and 2.40 eV. Raman scattering measurements were carried out using the same setup under 1.96 eV excitation. The laser was focused onto the sample using a Mitutoyo Plan Apo SL $50\times$ objective lens ($\text{NA} = 0.55$), producing a spot diameter of $\sim 1 \mu\text{m}$. The emitted signal was collected in a backscattering geometry through the same objective and directed to a Teledyne Princeton Instruments spectrometer equipped with a 150 grooves/mm grating and a CCD detector. The average excitation power was maintained at $\sim 30 \mu\text{W}$.

AFM Scan: AFM measurements were performed using a Dimension Icon microscope equipped with a NanoScope 6 controller (Bruker Corporation, Billerica, MA, USA). Topographical images were acquired in PeakForce Tapping mode using ScanAsyst-Air probes. The probes were calibrated using the thermal tuning method prior to measurements.

TRPL Measurements: TRPL investigations were performed using an S20 synchroscan streak camera system.

The samples were excited under 1.61 and 2.27 eV excitation using the MIRA laser system and the second harmonic of a Ti:sapphire fs oscillator. The average excitation power was kept at $\sim 26 \mu\text{W}$ with a spot diameter of $\sim 1 \mu\text{m}$.

V. AUTHOR CONTRIBUTIONS

A.K. conceived the project and provided the necessary funding. G., M.B., P.T., A.K.D., J.B., and A.W. prepared the hBN bubbles. G. and A.K. fabricated the samples. G. and M.R.M. performed the optical experiments. G., M.R., T.K., and P.K. carried out the TRPL experiments. G. carried out the data analysis with inputs taken from A.K. and M.R.M. M.A., D.D., and S.K.N. conducted the theoretical studies. G., A.K., M.R.M., and A.B. interpreted the experimental results. T.T. and K.W. provided the hBN crystals. G., A.K., and M.R.M. wrote the manuscript. All authors were consulted and their feedback was incorporated before submission.

VI. ACKNOWLEDGMENTS

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VII. COMPETING INTERESTS

We declare no competing financial interests.

VIII. SUPPORTING INFORMATION

Additional experimental details; AFM characterization, strain analysis from PL measurements; PL measurements under different excitation energies; relative PL quantum yield calculations; DFT calculations; TRPL measurements and analysis; and transfer-matrix calculations. The Supporting Information is available online.

IX. DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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